Hall Effect in Anisotropic Materials

October 13, 2026

1 Introduction

In an anisotropic material, the current density $\mathbf{J}$ and electric field $\mathbf{E}$ are related by a resistivity tensor ρ_{ij}. When a magnetic field is applied, the transverse electric field contains both an intrinsic anisotropic contribution and a magnetic-field-induced Hall contribution. For a two-dimensional system with current along direction 1 and transverse voltage along direction 2, the central result is

$$\boxed{\frac{E_2}{J_1} = \rho_{21}^{(0)} + R_{\mathrm{H}} B_3.} \tag{1}$$

Here $\rho_{21}^{(0)}$ is the zero-field transverse resistivity, R_{H} is the Hall coefficient, and B_3 is the magnetic field perpendicular to the plane.

2 Generalized Ohm's Law in Anisotropic Media

For an anisotropic medium, the generalized Ohm's law is

$$E_i = \sum_j \rho_{ij} J_j, \tag{2}$$

where ρ_{ij} is the resistivity tensor. In a thin-film geometry, if the current is applied along direction 1 and the transverse current is zero, $J_2 = 0$, then

$$E_2 = \rho_{21} J_1. \tag{3}$$

Thus the measured transverse resistivity is simply

$$\frac{E_2}{J_1} = \rho_{21}. \tag{4}$$

In an isotropic material, $\rho_{21}^{(0)} = 0$ at zero magnetic field. In an anisotropic material, $\rho_{21}^{(0)}$ can be nonzero even without a magnetic field. This is the origin of the planar Hall effect and anisotropic magnetoresistance [5, 6].

3 Magnetic-Field Expansion and Onsager Relations

For a weak magnetic field $\mathbf{B}$, the resistivity tensor can be expanded as

$$\rho_{ij}(\mathbf{B}) = \rho_{ij}^{(0)} + \sum_k R_{ijk} B_k + \mathcal{O}(B^2), \tag{5}$$

where $\rho_{ij}^{(0)}$ is the zero-field resistivity tensor and R_{ijk} is a third-rank tensor describing the linear magnetoresistance and Hall effect [1–3].

The Onsager reciprocal relations require

$$\rho_{ij}(\mathbf{B}) = \rho_{ji}(-\mathbf{B}). \tag{6}$$

For a magnetic field along direction 3, $\mathbf{B} = B_3\hat{z}$, the transverse component becomes

$$\rho_{21}(B_3) = \rho_{21}^{(0)} + R_{213}B_3. \tag{7}$$

Defining the Hall coefficient

$$R_{\mathrm{H}} \equiv R_{213}, \tag{8}$$

we obtain

$$\rho_{21}(B_3) = \rho_{21}^{(0)} + R_{\mathrm{H}}B_3. \tag{9}$$

4 Derivation of the Transverse Resistivity

Using $E_2 = \rho_{21}J_1$, we divide by J_1:

$$\frac{E_2}{J_1} = \rho_{21}(B_3) = \rho_{21}^{(0)} + R_{\mathrm{H}}B_3. \tag{10}$$

This is the desired relation. The first term is independent of the magnetic field and arises from the intrinsic anisotropy of the material. The second term is linear in B_3 and is the ordinary Hall contribution. In simple free-electron models, $R_{\mathrm{H}} = -1/(ne)$, where n is the carrier density and e is the elementary charge [1, 2]. In ferromagnetic or topological materials, additional anomalous or planar Hall contributions may appear [7].

5 Physical Interpretation

- $\rho_{21}^{(0)}$: zero-field transverse resistivity. It is nonzero only when the material is anisotropic and the current is not along a principal axis. It is even in $\mathbf{B}$.
- $R_{\mathrm{H}}B_3$: ordinary Hall term. It is odd in $\mathbf{B}$ and changes sign when the magnetic field is reversed.

Therefore, the total transverse response is a superposition of an anisotropic, field-independent part and a Hall part linear in the magnetic field.

References

[1] N. W. Ashcroft and N. D. Mermin, *Solid State Physics*, Holt, Rinehart and Winston, New York, 1976.

[2] C. Kittel, *Introduction to Solid State Physics*, 8th ed., John Wiley & Sons, Hoboken, 2005.

[3] L. D. Landau, E. M. Lifshitz, and L. P. Pitaevskii, *Electrodynamics of Continuous Media*, 2nd ed., Pergamon, Oxford, 1984.

[4] L. Onsager, "Reciprocal Relations in Irreversible Processes. I.," *Physical Review* **37**, 405–426 (1931).

[5] T. R. McGuire and R. I. Potter, "Anisotropic Magnetoresistance in Ferromagnetic 3d Alloys," *IEEE Transactions on Magnetics* **11**, 1018–1038 (1975).

[6] R. C. O'Handley, *Modern Magnetic Materials: Principles and Applications*, Wiley, New York, 2000.

[7] N. Nagaosa, J. Sinova, S. Onoda, A. H. MacDonald, and N. P. Ong, "Anomalous Hall Effect," *Reviews of Modern Physics* **82**, 1539–1592 (2010).